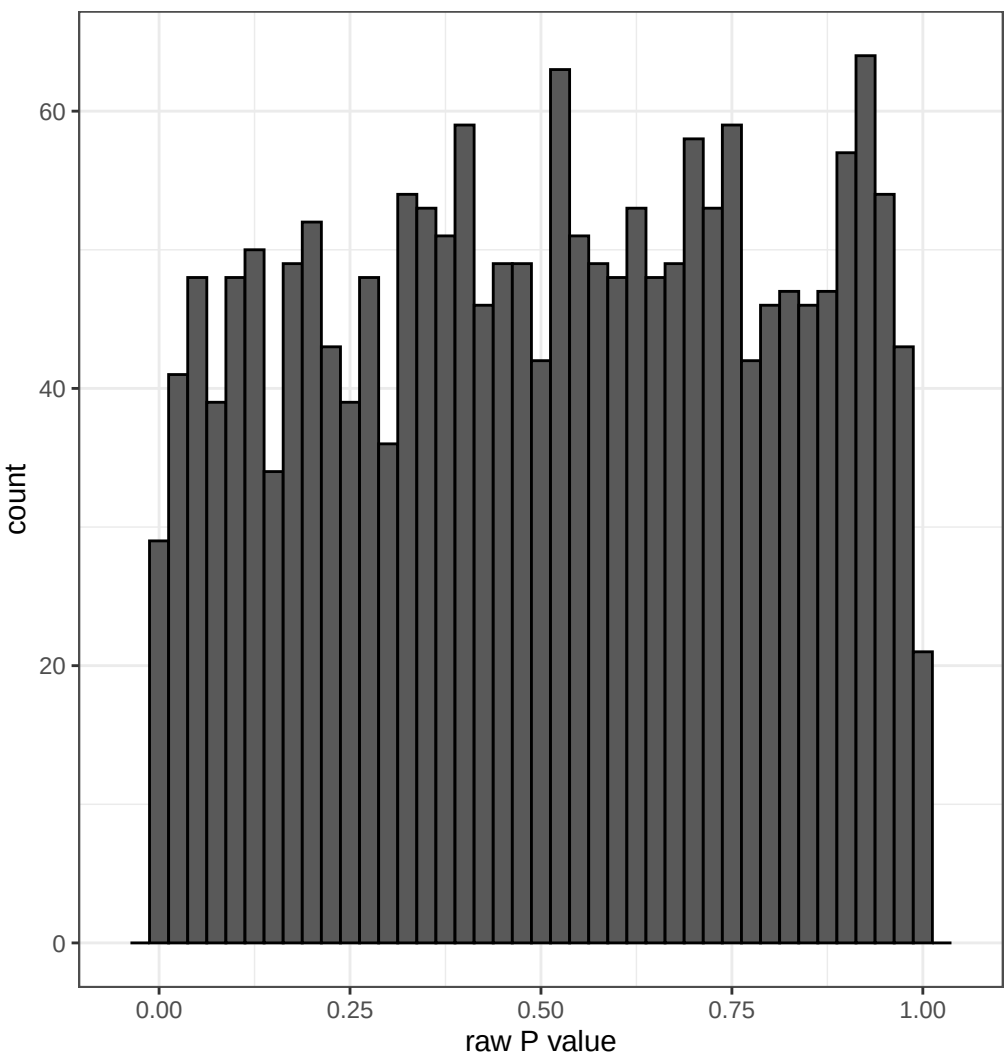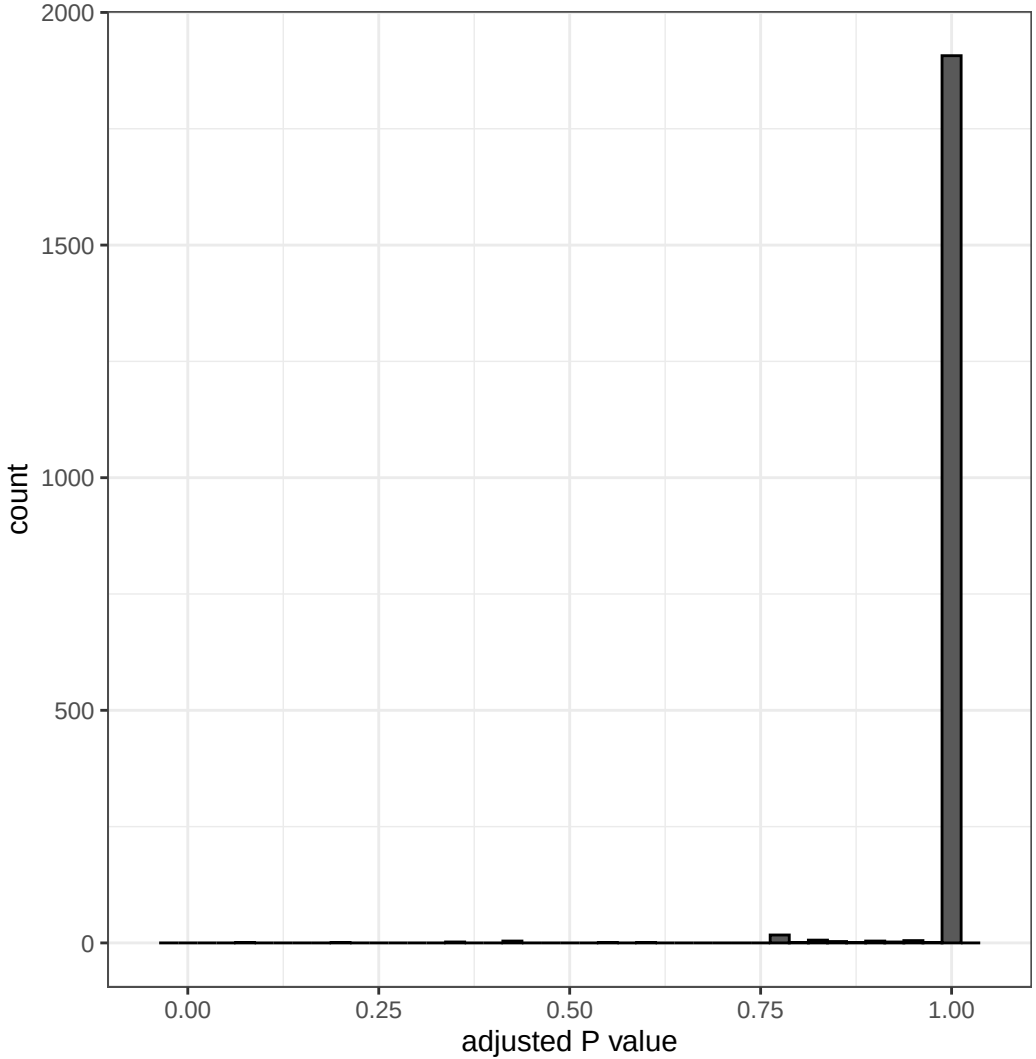Interaction_supp